



King Abdul Aziz University

Faculty of Computing & Information Technology

Computer Science Department



CPCS 212 - APPLIED MATH FOR COMPUTING

Nonlinear Equation Bisection Method

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Numerical Methods

- **Numerical Methods:**

Algorithms that are used to obtain numerical solutions of a mathematical problem.

- **Why do we need them?**

1. No analytical solution exists,
2. An analytical solution is difficult to obtain or not practical.

What do we need?

Basic Needs in the Numerical Methods:

- **Practical:**
 - Can be computed in a reasonable amount of time.
- **Accurate:**
 - Good approximate to the true value,
 - Information about the approximation error
(Bounds, error order,...)

Solution of Nonlinear Equations

- Some simple equations can be solved analytically:

$$x^2 + 4x + 3 = 0$$

$$\text{Analytic solution roots} = \frac{-4 \pm \sqrt{4^2 - 4(1)(3)}}{2(1)}$$

$$x = -1 \text{ and } x = -3$$

- Many other equations have no analytical solution:

$$x^9 - 2x^2 + 5 = 0$$

$$x = e^{-x}$$

No analytic solution

Methods for Solving Nonlinear Equations

- Bisection Method
- Newton-Raphson Method
- Secant Method

Solution of Nonlinear Equations (Root Finding Problems)

Definitions

Classification of Methods

- Analytical Solutions
- Graphical Methods
- Numerical Methods
 - Bracketing Methods
 - Open Methods

Convergence Notations

Root Finding Problems

Many problems in Science and Engineering are expressed as:

Given a continuous function $f(x)$,
find the value r such that $f(r) = 0$

These problems are called root finding problems.

Roots of Equations

A number r that satisfies an equation is called a root of the equation.

The equation: $x^4 - 3x^3 - 7x^2 + 15x = -18$

has four roots: $-2, 3, 3, \text{and } -1$.

i.e., $x^4 - 3x^3 - 7x^2 + 15x + 18 = (x + 2)(x - 3)^2(x + 1)$

The equation has two simple roots (-1 and -2) and a repeated root (3) with multiplicity $= 2$.

Zeros of a Function

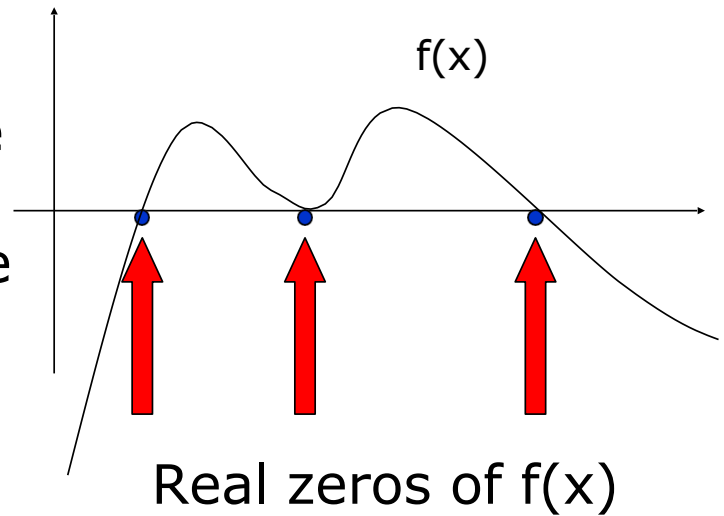
Let $f(x)$ be a real-valued function of a real variable. Any number r for which $f(r)=0$ is called a zero of the function.

Examples:

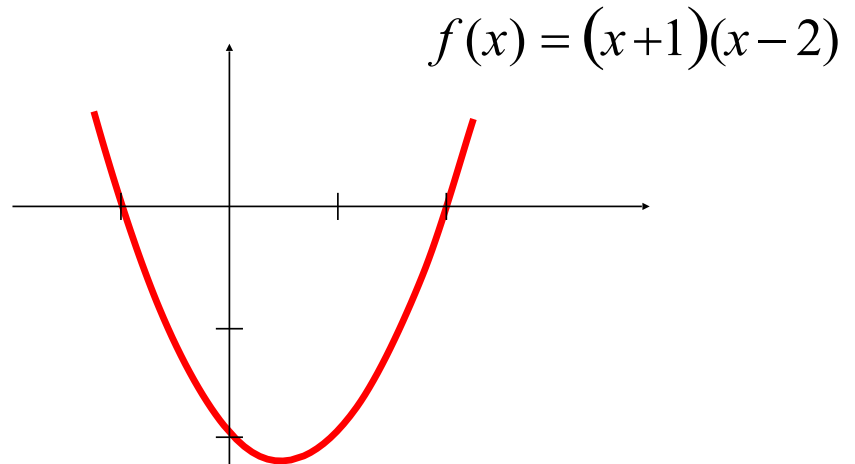
2 and 3 are zeros of the function $f(x) = (x-2)(x-3)$.

Graphical Interpretation of Zeros

- The real zeros of a function $f(x)$ are the values of x at which the graph of the function crosses (or touches) the x -axis.



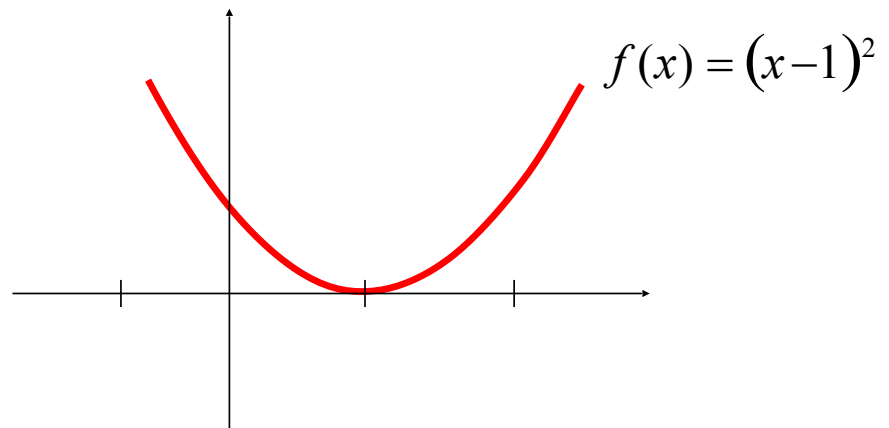
Simple Zeros



$$f(x) = (x+1)(x-2) = x^2 - x - 2$$

has two simple zeros (one at $x = 2$ and one at $x = -1$)

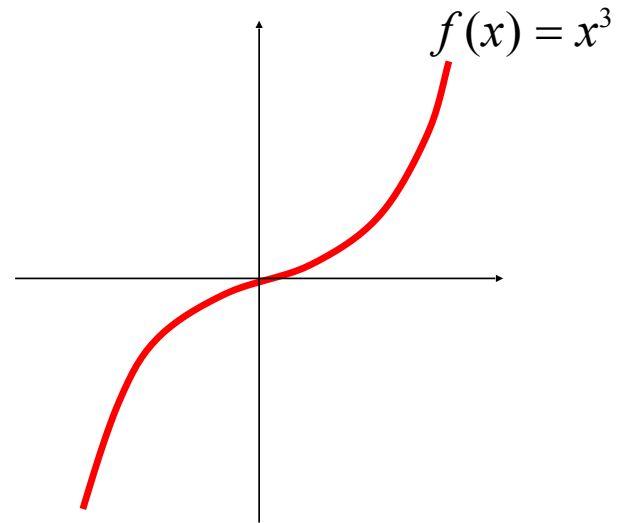
Multiple Zeros



$$f(x) = (x-1)^2 = x^2 - 2x + 1$$

has double zeros (zero with multiplicity = 2) at $x = 1$

Multiple Zeros



$$f(x) = x^3$$

has a zero with multiplicity = 3 at $x = 0$

Facts

- Any n^{th} order polynomial has exactly n zeros (counting real and complex zeros with their multiplicities).
- Any polynomial with an odd order has at least one real zero.
- If a function has a zero at $\mathbf{x=r}$ with multiplicity **m** then the function and its first **$(m-1)$** derivatives are zero at $\mathbf{x=r}$ and the **m^{th} derivative at $\mathbf{r}$ is not zero.**

Roots of Equations & Zeros of Function

Given the equation :

$$x^4 - 3x^3 - 7x^2 + 15x = -18$$

Move all terms to one side of the equation :

$$x^4 - 3x^3 - 7x^2 + 15x + 18 = 0$$

Define $f(x)$ as :

$$f(x) = x^4 - 3x^3 - 7x^2 + 15x + 18$$

The zeros of $f(x)$ are the same as the roots of the equation $f(x) = 0$
(Which are $-2, 3, 3$, and -1)

Solution Methods

Several ways to solve nonlinear equations are possible:

- Analytical Solutions
 - Possible for special equations only
- Graphical Solutions
 - Useful for providing initial guesses for other methods
- Numerical Solutions
 - Open methods
 - Bracketing methods

Analytical Methods

Analytical Solutions are available for special equations only.

Analytical solution of : $a x^2 + b x + c = 0$

$$roots = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

No analytical solution is available for : $x - e^{-x} = 0$

Graphical Methods

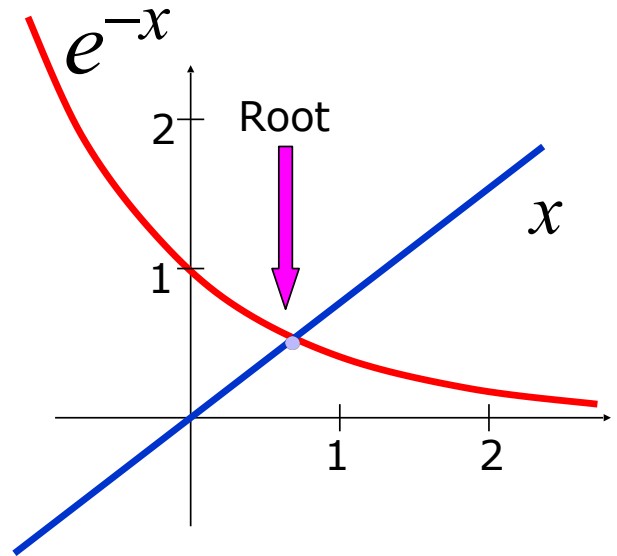
- Graphical methods are useful to provide an initial guess to be used by other methods.

Solve

$$x = e^{-x}$$

The root $\in [0,1]$

root ≈ 0.6



Numerical Methods

Many methods are available to solve nonlinear equations:

- Bisection Method
- Newton's Method
- Secant Method
- False position Method
- Muller's Method
- Bairstow's Method
- Fixed point iterations

Bisection Method

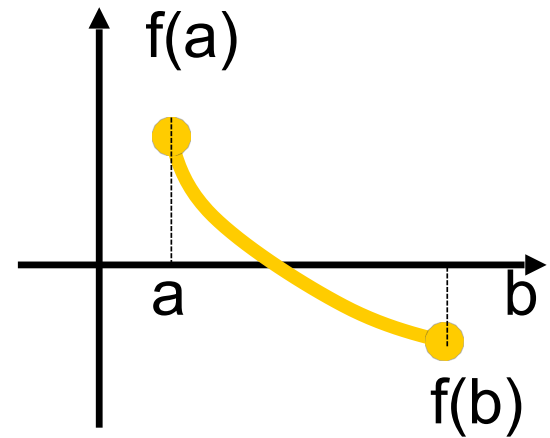
- The Bisection Algorithm
- Convergence Analysis of Bisection
- Method Examples

Introduction

- ❑ The **Bisection method** is one of the simplest methods to find a zero of a nonlinear function.
- ❑ It is also called **interval halving** method.
- ❑ To use the Bisection method, one needs an initial interval that is known to contain a zero of the function.
- ❑ The method systematically reduces the interval. It does this by dividing the interval into two equal parts, performs a simple test and based on the result of the test, half of the interval is thrown away.
- ❑ The procedure is repeated until the desired interval size is obtained.

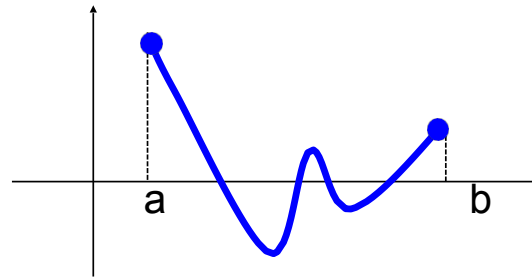
Intermediate Value Theorem

- Let $f(x)$ be defined on the interval $[a,b]$.
- Intermediate value theorem:
if a function is continuous and $f(a)$ and $f(b)$ have different signs then the function has at least one zero in the interval $[a,b]$.



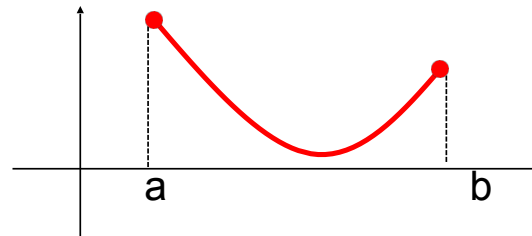
Examples

- If $f(a)$ and $f(b)$ have the same sign, the function may have an even number of real zeros or no real zeros in the interval $[a, b]$.



The function has four real zeros

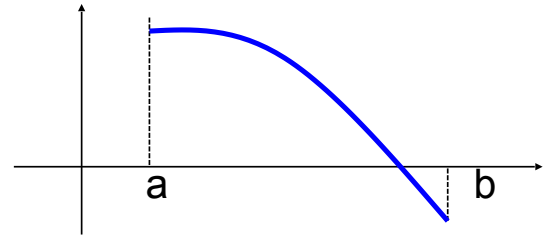
- Bisection method can not be used in these cases.



The function has no real zeros

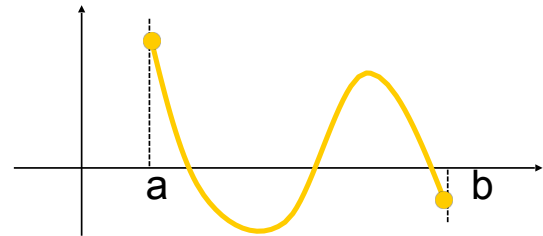
Two More Examples

▣ If $f(a)$ and $f(b)$ have different signs, the function has at least one real zero.



The function has one real zero

▣ Bisection method can be used to find one of the zeros.



The function has three real zeros

Bisection Method

- If the function is continuous on $[a,b]$ and $f(a)$ and $f(b)$ have different signs, Bisection method obtains a new interval that is half of the current interval and the sign of the function at the end points of the interval are different.
- This allows us to repeat the Bisection procedure to further reduce the size of the interval.

Bisection Method

Assumptions:

Given an interval $[a, b]$

$f(x)$ is continuous on $[a, b]$

$f(a)$ and $f(b)$ have opposite signs.

These assumptions ensure the existence of at least one zero in the interval $[a, b]$ and the bisection method can be used to obtain a smaller interval that contains the zero.

Bisection Algorithm

Assumptions:

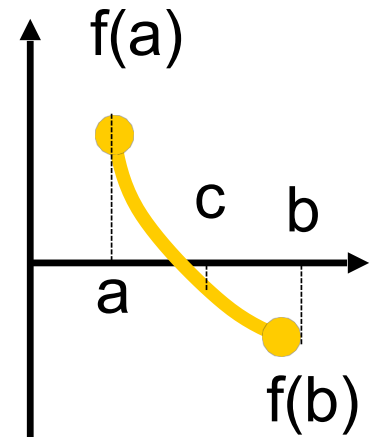
- ▣ $f(x)$ is continuous on $[a, b]$
- ▣ $f(a) f(b) < 0$

Algorithm:

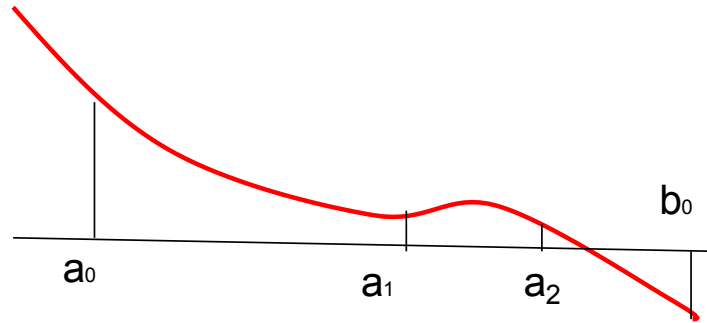
Loop

1. Compute the mid point $c = (a+b)/2$
2. Evaluate $f(c)$
3. If $f(a) f(c) < 0$ then new interval $[a, c]$
If $f(a) f(c) > 0$ then new interval $[c, b]$

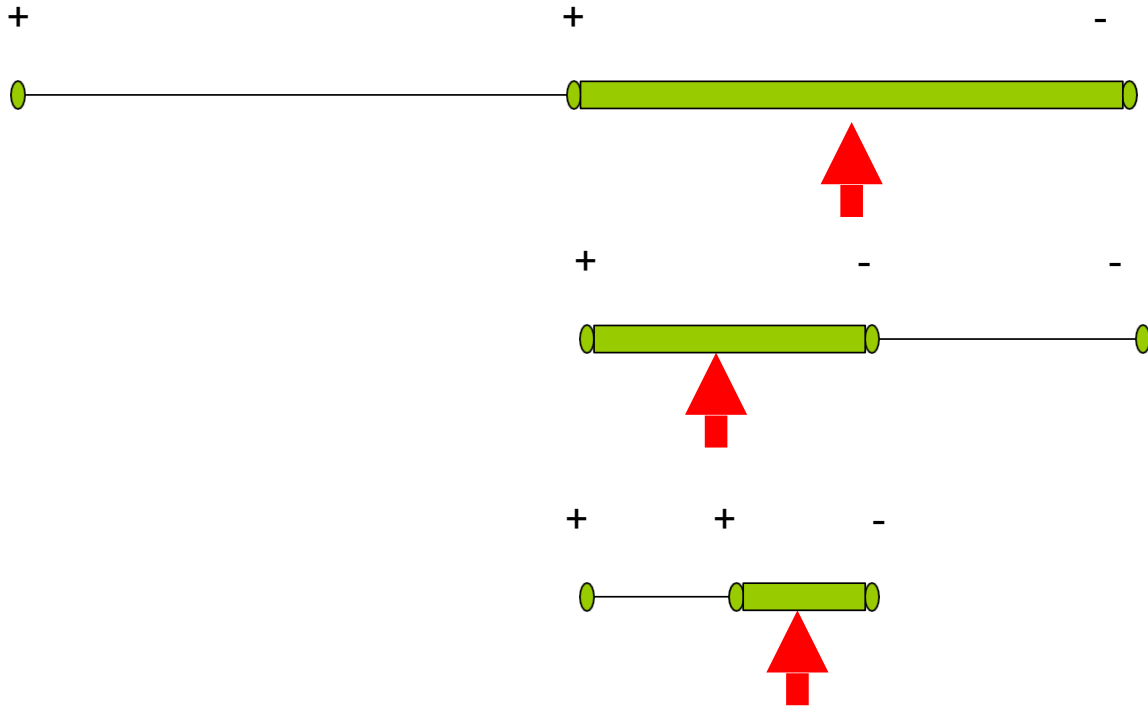
End loop



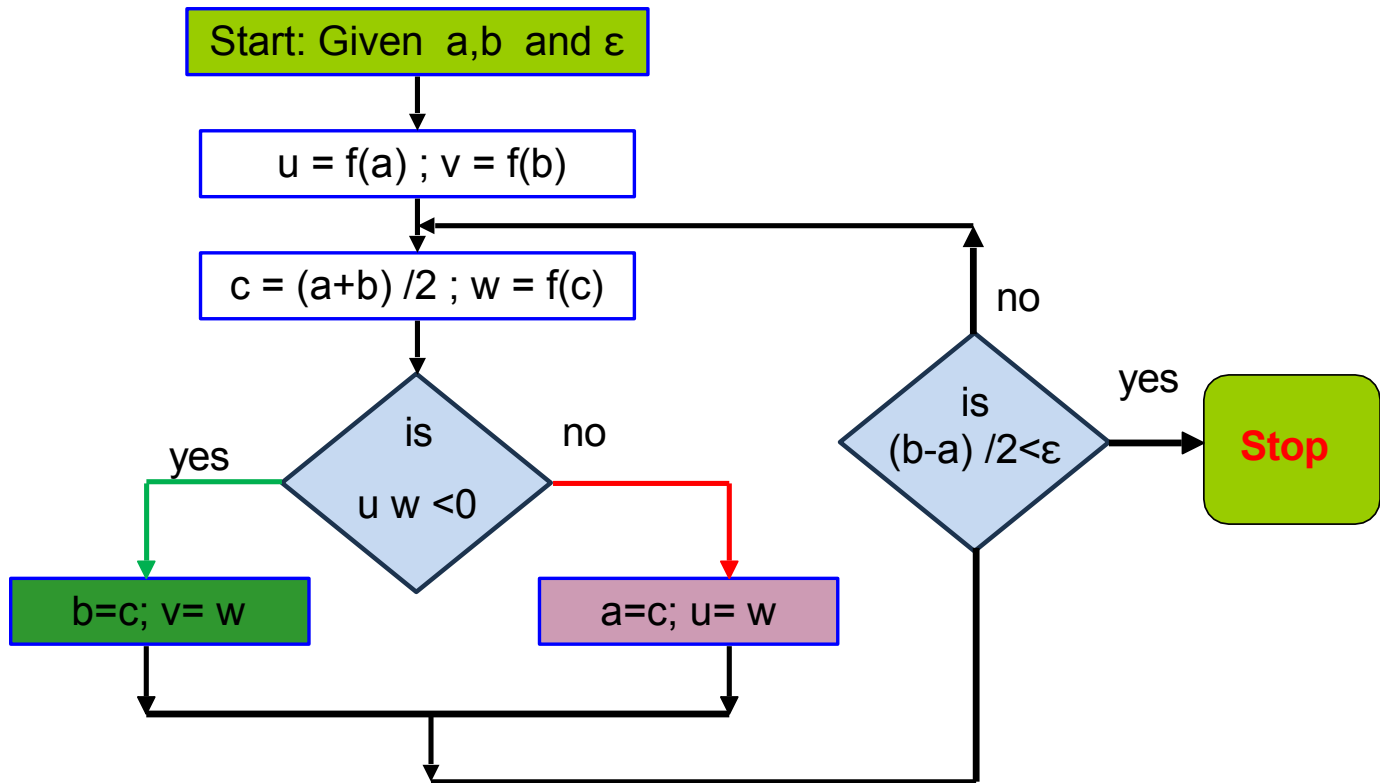
Bisection Method



Example



Flow Chart of Bisection Method



Example

Can you use Bisection method to find a zero of :

$$f(x) = x^3 - 3x + 1 \quad \text{in the interval } [0,2]?$$

Answer:

$f(x)$ is continuous on $[0,2]$

$$\text{and } f(0) * f(2) = (1)(3) = 3 > 0$$

$\Rightarrow$ Assumptions are not satisfied

$\Rightarrow$ Bisection method can not be used

Example

Can you use Bisection method to find a zero of :

$f(x) = x^3 - 3x + 1$ in the interval $[0,1]$?

Answer:

$f(x)$ is continuous on $[0,1]$

and $f(0) * f(1) = (1)(-1) = -1 < 0$

$\Rightarrow$ Assumptions are satisfied

$\Rightarrow$ Bisection method can be used

Best Estimate and Error Level

Bisection method obtains an interval that is guaranteed to contain a zero of the function.

Questions:

- ▣ What is the best estimate of the zero of $f(x)$?
- ▣ What is the error level in the obtained estimate?

Best Estimate and Error Level

The best estimate of the zero of the function **$f(\mathbf{x})$** after the first iteration of the Bisection method is the mid point of the initial interval:

$$\textit{Estimate of the zero: } r = \frac{b+a}{2}$$

$$\textit{Error} \leq \frac{b-a}{2}$$

Stopping Criteria

Two common stopping criteria:

- Stop after a fixed number of iterations
- Stop when the absolute error is less than a specified value

How are these criteria related?

Stopping Criteria

- c_n : is the midpoint of the interval at the n^{th} iteration
(c_n is usually used as the estimate of the root).
- r : is the zero of the function.

After n iterations :

$$|error| = |r - c_n| \leq E_a^n = \frac{b-a}{2^n} = \frac{\Delta x^0}{2^n}$$

Convergence Analysis

Given $f(x)$, a , b , and ε

How many iterations are needed such that: $|x - r| \leq \varepsilon$
where r is the zero of $f(x)$ and x is the
bisection estimate (i.e., $x = c_k$)?

$$n \geq \frac{\log(b - a) - \log(\varepsilon)}{\log(2)}$$

Convergence Analysis – Alternative Form

$$n \geq \frac{\log(b - a) - \log(\varepsilon)}{\log(2)}$$

$$n \geq \log_2 \left(\frac{\text{width of initial interval}}{\text{desired error}} \right) = \log_2 \left(\frac{b - a}{\varepsilon} \right)$$

Example 1

$$a = 6, b = 7, \varepsilon = 0.0005$$

How many iterations are needed such that: $|x - r| \leq \varepsilon$?

$$n \geq \frac{\log(b-a) - \log(\varepsilon)}{\log(2)} = \frac{\log(1) - \log(0.0005)}{\log(2)} = 10.9658$$

$$\Rightarrow n \geq 11$$

Example

- Use Bisection method to find a root of the equation $x = \cos(x)$ with absolute error < 0.02 (assume the initial interval $[0.5, 0.9]$)

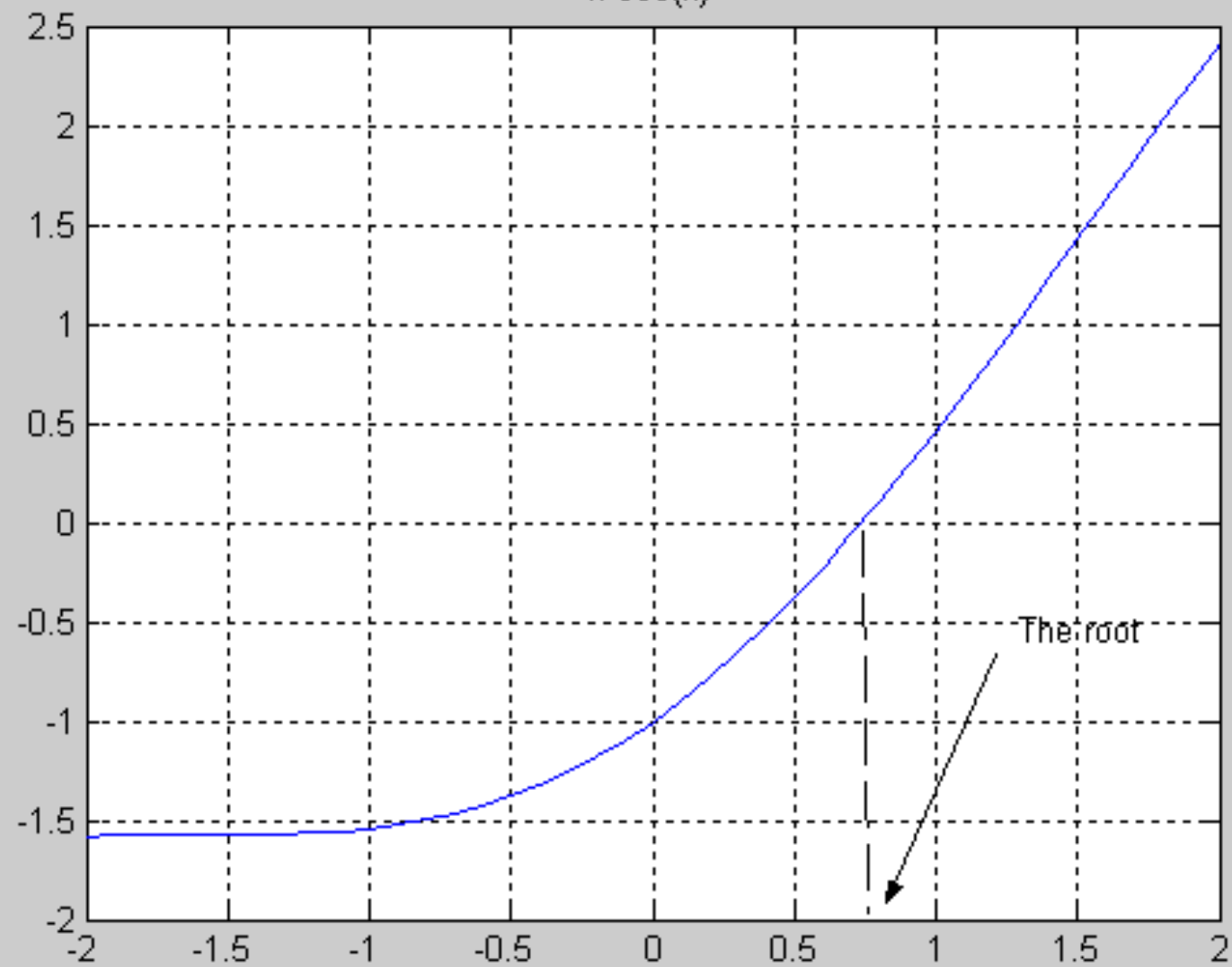
Question 1: What is $f(x)$?

Question 2: Are the assumptions satisfied ?

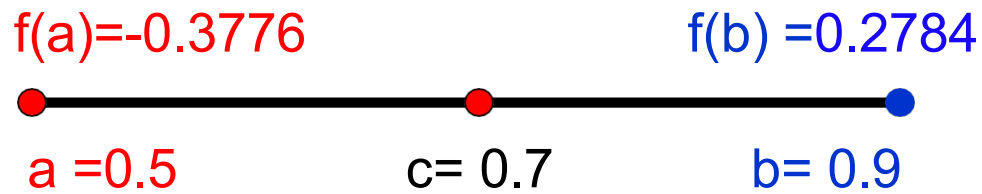
Question 3: How many iterations are needed ?

Question 4: How to compute the new estimate ?

$$x - \cos(x)$$

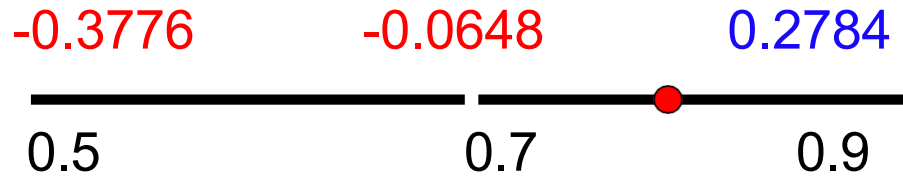


Example 1 Initial Interval

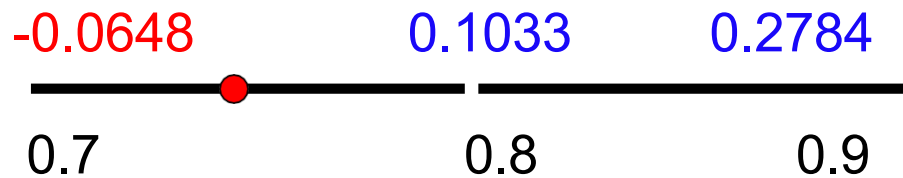


Error < 0.2

Example 1

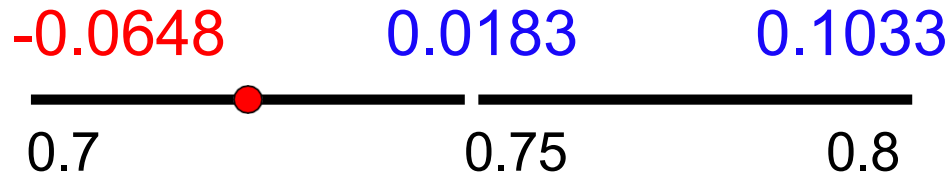


Error < 0.1

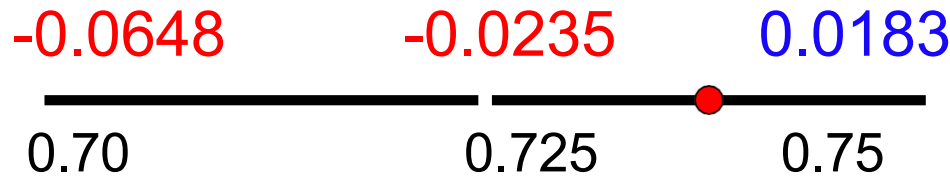


Error < 0.05

Example 1



Error < 0.025



Error < .0125

Example 1 Summary

- ▣ Initial interval containing the root:
 $[0.5, 0.9]$

- ▣ After 5 iterations:
 - Interval containing the root: $[0.725, 0.75]$
 - Best estimate of the root is 0.7375
 - $|\text{Error}| < 0.0125$

Matlab Program of Bisection Method

```
a=.5; b=.9;  
u=a-cos(a);  
v=b-cos(b);  
for i=1:5  
    c=(a+b)/2  
    fc=c-cos(c)  
    if u*fc<0  
        b=c ; v=fc;  
    else  
        a=c; u=fc;  
    end  
end
```

```
c =  
    0.7000  
fc =  
   -0.0648  
c =  
    0.8000  
fc =  
    0.1033  
c =  
    0.7500  
fc =  
    0.0183  
c =  
    0.7250  
fc =  
   -0.0235
```

Example 2

Find the root of:

$$f(x) = x^3 - 3x + 1 \text{ in the interval: } [0, 1]$$

* $f(x)$ is continuous

* $f(0) = 1, f(1) = -1 \Rightarrow f(a) f(b) < 0$

$\Rightarrow$ Bisection method can be used to find the root

Example 2

Iteration	a	b	$c = \frac{(a+b)}{2}$	f(c)	$\frac{(b-a)}{2}$
1	0	1	0.5	-0.375	0.5
2	0	0.5	0.25	0.266	0.25
3	0.25	0.5	.375	-7.23E-3	0.125
4	0.25	0.375	0.3125	9.30E-2	0.0625
5	0.3125	0.375	0.34375	9.37E-3	0.03125

Example 3

- Perform 3 iterations of the bisection method on the function $f(x) = x^3 - 4$, with starting interval $[1, 3]$.

Bisection Method

Advantages

- **Simple** and easy to implement
- **One** function evaluation per iteration
- The **size** of the interval containing the zero is reduced by 50% after each iteration
- The **number of iterations** can be determined **a priori** No knowledge of the **derivative** is needed
- The function does **not** have to be **differentiable**

Disadvantage

- **Slow** to converge
- **Good** intermediate approximations may be **discarded**